Eric Du

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EXPERIENCE

Software Engineer Apr 2024 – Aug 2026
Symmetry Laser | Davis, CA
Only software engineer at a 50-person company, reporting to the CTO. Chose what to build and ran it in production.

- Built a **200GB** PostgreSQL replica of the core FileMaker database and moved 12+ services onto it, cutting p50 reads from **4s to 10ms**. Later used it to recover all **8 hours** of writes lost when a failed update corrupted the primary.
- Replaced FileMaker's JDBC and ODBC drivers company-wide with a Rust reimplementation of its undocumented wire protocol, adding connection recovery, tracing, and table-affinity pooling that cut p50 latency **4x**.
- Took 8 services from hand-deployed VMs, where a crash stayed down until someone noticed, to Docker and Gitea CI/CD, reaching **99.9%** health-checked uptime and cutting deploys from **20 minutes to 1**.
- Wrote a service that precompiles CNC programs and streams them over TLS to 19 laser controllers at two sites, cutting worst-case load times from **4 minutes to 100ms**, with checksums and network-drive fallback.
- Built an AWS Bedrock LLM assistant, piloted with sales reps, that queries company data through read-only Postgres roles scoped to fixed views on the replica, so the database, not the prompt, limits what the model can read.
- Built VS Code extensions with completion, hover docs, and live compile errors for two proprietary CNC languages. All 14 engineers adopted them, and floor compile failures went from **5 a week to zero** for 5 months.

Full Stack Engineer Apr 2023 – Apr 2024
UC Davis, Department of Plant Sciences | Davis, CA

- Built a FastAPI and AWS Lambda service that predicts planting dates from weather-station data for **30+** researchers.
- Shipped fuzzy search over **60,000+** research papers to the department's public site, on MySQL and AWS.

Full Stack Engineer Sep 2022 – Dec 2023
AggieWorks | Davis, CA
Student-run software team building products for the UC Davis community.

- Built weighted compatibility scoring over 23 criteria for RoomU, a React Native and Go roommate app with **120+ users**.
- Built full-stack WebSocket messaging with paginated history for AggieExchange, a Next.js marketplace with **160+ users**, and cut image p90 from **500ms to 100ms** by downscaling and caching uploads.

PROJECTS

Costco Gas Tracker | React, FastAPI, Redis, TimescaleDB, Docker | github.com/EricDooo/costco-pump Aug 2026

- Tracks live fuel prices at every Costco station across **11 countries**, reconciling three undocumented APIs across two Costco platforms into one feed with full price history in a TimescaleDB hypertable.
- Split ingestion into independent Redis/RQ jobs so one slow country never blocks the rest, and self-hosted the Protomaps basemap to run with **no third-party map API key or quota**.
- Deployed with Docker Compose, GitHub Actions, and GHCR to an arm64 VM behind Caddy, and blocked dependency versions less than 14 days old to guard against supply-chain attacks.

TECHNICAL SKILLS

Languages: Rust, Python, TypeScript, JavaScript, C#, Go, SQL, Bash
Backend: FastAPI, Axum, Django, Node.js, REST APIs, WebSockets
Data: PostgreSQL, TimescaleDB, PostGIS, Redis, MySQL, InfluxDB, Apache Airflow
Infrastructure: Docker, GitHub Actions, CI/CD, Linux, Git, AWS (Lambda, S3, Bedrock)
Observability and Testing: OpenTelemetry, Grafana, Loki, pytest, cargo test, Playwright
Frontend: React, SvelteKit, Vue.js, Next.js, React Native

EDUCATION

University of California, Davis | Davis, CA Sep 2020 – Dec 2023
Bachelor of Science in Computer Science